

THE MISSING RATIO: $\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$ AS A FUNDAMENTAL CONSTANT

ABSTRACT. The Euler reflection formula at $z = \frac{1}{4}$ gives $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$. This *product* is classical and well studied. We observe that the *ratio* $\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4}) \approx 2.9587$ —which we denote G^* and call the *lemniscatic bridge constant*—has received almost no independent attention, despite admitting a Wallis-type infinite product, appearing as the scaled period of the CM elliptic curve $y^2 = x^3 - x$, and factoring the lemniscate constant as $\varpi = G^*\sqrt{\pi}/2$.

We argue that G^* deserves recognition as a fundamental constant on the same footing as π and ϖ , and that its historical absence from the literature is an accident of convention, not of substance.

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1. THE REFLECTION FORMULA PRODUCES TWO OBJECTS

The Euler reflection formula states that for all $z \notin \mathbb{Z}$,

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}. \quad (1)$$

At $z = \frac{1}{4}$:

$$\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2}. \quad (2)$$

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This is the *product*. It yields a circular quantity (π , modified by $\sqrt{2}$) and has been known since Euler. It appears in hundreds of papers and textbooks.

The same two Gamma values also determine a *ratio*:

$$\frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})} = \frac{\Gamma(\frac{1}{4})^2}{\pi\sqrt{2}} = 2.9586751191886388\dots \quad (3)$$

This number, which we denote G^* , has—to our knowledge—never been given a standard name or symbol, despite being as elementary as the product.

Remark 1.1. Every evaluation of the reflection formula at a rational argument $z = p/q$ produces a product and a ratio. At $z = \frac{1}{2}$, the product $\Gamma(\frac{1}{2})^2 = \pi$ yields the circle constant itself, and the ratio $\Gamma(\frac{1}{2})/\Gamma(\frac{1}{2}) = 1$ is trivial. At $z = \frac{1}{4}$, the product yields $\pi\sqrt{2}$ and the ratio yields G^* . This is the *first* non-trivial ratio produced by the reflection formula at a point of the form $z = 1/(2n)$.

2. WHY THE RATIO WAS OVERLOOKED

We identify three historical reasons why G^* was not recognised as a fundamental constant.

2.1. The tradition of periods. Since Euler, the preferred objects in elliptic function theory have been *periods*—the arc lengths, areas, and integrals that arise as lattice generators. The lemniscate constant

$$\varpi = \frac{\Gamma(\frac{1}{4})^2}{2\sqrt{2}\Gamma(\frac{1}{2})} = 2.6220575542921120\dots \quad (4)$$

is the half-perimeter of the lemniscate $r^2 = \cos 2\theta$ and a period of the elliptic curve $y^2 = x^3 - x$. As a period, it has geometric meaning: it is a length one can draw. The constant G^* is not directly a period; it is a *ratio* of two Gamma values. Periods were named and studied; ratios were computed and discarded.

2.2. The dominance of π . The reflection formula (2) was primarily used to compute π and its relatives, not to study the Gamma function's internal structure. The product $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$ was valued because it gave π ; the ratio was an intermediate quantity with no known application.

2.3. The π -contaminated notation. The standard representation $\varpi = \Gamma(\frac{1}{4})^2/(2\sqrt{2}\pi)$ merges $\Gamma(\frac{1}{4})$ and π into a single expression, hiding the factorisation $\varpi = G^*\sqrt{\pi}/2$. In this notation, G^* is invisible: it appears only as a ratio of two subexpressions of ϖ that are never separated. The notation discouraged the decomposition.

3. THE CONSTANT AND ITS PROPERTIES

Definition 3.1. The *lemniscatic bridge constant* is

$$G^* = \frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})} = 2.9586751191886388\dots \quad (5)$$

We collect its principal representations, all classical:

$$G^* = \frac{\Gamma(\frac{1}{4})^2}{\sqrt{2}\Gamma(\frac{1}{2})^2} \quad (\pi\text{-free form}) \quad (6)$$

$$G^* = \frac{2\varpi}{\sqrt{\pi}} \quad (\text{lemniscate form}) \quad (7)$$

$$G^* = \frac{2\sqrt{2}}{\sqrt{\pi}} K\left(\frac{1}{\sqrt{2}}\right) \quad (\text{elliptic integral}) \quad (8)$$

$$G^* = \frac{2\sqrt{\pi}}{\text{agm}(1, \sqrt{2})} \quad (\text{AGM}) \quad (9)$$

$$G^* = \sqrt{2\pi} \theta_3(e^{-\pi})^2 \quad (\text{theta function at self-dual nome}) \quad (10)$$

$$G^* = \frac{8L(E, 1)}{\sqrt{\pi}} \quad (L\text{-function of } y^2 = x^3 - x) \quad (11)$$

Each representation is known; none was previously attributed to a named constant.

4. WHAT THE RATIO KNOWS THAT THE PRODUCT DOES NOT

The product $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})$ is *symmetric*: it treats $z = \frac{1}{4}$ and $z = \frac{3}{4}$ on equal footing. Swapping them changes nothing. The product yields $\pi\sqrt{2}$, a quantity that depends only on the *sum* $z + (1 - z) = 1$, not on the specific partition into $\frac{1}{4}$ and $\frac{3}{4}$.

The ratio $\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$ is *asymmetric*: it distinguishes $z = \frac{1}{4}$ from $z = \frac{3}{4}$. It measures *how much larger* $\Gamma(\frac{1}{4})$ is than $\Gamma(\frac{3}{4})$. This asymmetry carries information:

Theorem 4.1 (Wallis product for the ratio).

$$G^* = \lim_{N \rightarrow \infty} (N+1)^{-1/2} \prod_{k=0}^N \frac{4k+3}{4k+1}. \quad (12)$$

Proof. Write $Q_N = \prod_{k=0}^N (4k+3)/(4k+1) = (\frac{3}{4})_{N+1}/(\frac{1}{4})_{N+1} = \Gamma(\frac{3}{4} + N + 1)\Gamma(\frac{1}{4})/[\Gamma(\frac{1}{4} + N + 1)\Gamma(\frac{3}{4})]$. By the asymptotic ratio $\Gamma(x+a)/\Gamma(x+b) \sim x^{a-b}$, $Q_N \sim [\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})](N+1)^{1/2}$. $\square$

The numerators 3, 7, 11, 15, 19, 23, ... are integers $\equiv 3 \pmod{4}$. The denominators 1, 5, 9, 13, 17, 21, ... are integers $\equiv 1 \pmod{4}$. In the arithmetic of the Gaussian integers $\mathbb{Z}[i]$, the former are *inert* (they remain prime) and the latter include the *split* primes (they factor as $\pi\bar{\pi}$). The ratio measures the cumulative advantage of inertness over splitting—the Fermat two-square classification made multiplicative.

The product knows about π . The ratio knows about the lattice.

5. THE TRIAD AND THE COMPOSITE NATURE OF ϖ

Theorem 5.1 (Factorisation of ϖ).

$$\varpi = \frac{G^* \cdot \sqrt{\pi}}{2}. \quad (13)$$

Proof. $G^* \sqrt{\pi}/2 = [\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})] \cdot \Gamma(\frac{1}{2})/2$. By reflection, $\Gamma(\frac{3}{4}) = \pi\sqrt{2}/\Gamma(\frac{1}{4})$, so $G^* \sqrt{\pi}/2 = \Gamma(\frac{1}{4})^2 \Gamma(\frac{1}{2}) / (2\pi\sqrt{2}) = \Gamma(\frac{1}{4})^2 / (2\sqrt{2}\Gamma(\frac{1}{2})) = \varpi$. $\square$

The lemniscate constant, treated since Euler as a single irreducible period, factors into two independent pieces:

- $\sqrt{\pi}$, which arises from the Wallis product over the even/odd partition (the mod-2 “race”);
- G^* , which arises from the Wallis product over the 1 mod 4 / 3 mod 4 partition (the mod-4 “race”).

Neither piece alone yields ϖ ; both are required.

Proposition 5.2 (No single-product representation of ϖ). *There is no ratio product $\prod (an + b)/(an + c)$ over a single arithmetic progression that converges (after polynomial prefactor) to ϖ .*

Argument. Such a product converges to a ratio $\Gamma(c/a)/\Gamma(b/a)$ (times algebraic factors), which involves $\Gamma(\frac{1}{4})$ at most once. The lemniscate constant requires $\Gamma(\frac{1}{4})^2$ in the numerator, which cannot be produced by a single ratio. $\square$

6. THE TRIAD IDENTITY

Theorem 6.1. $\pi = 4\varpi^2/G^{*2}$.

Proof. $4\varpi^2/G^{*2} = 4 \cdot (G^* \sqrt{\pi}/2)^2 / G^{*2} = 4 \cdot G^{*2} \pi / (4G^{*2}) = \pi$. $\square$

This expresses the circle constant as a ratio of two lemniscatic objects. In the hierarchy $\Gamma(\frac{1}{4}) \rightarrow G^* \rightarrow \varpi \rightarrow \pi$, the ratio precedes the period, and the period precedes the circle.

7. WHY THIS MATTERS

The constant $G^* = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$ appears identically in at least seven branches of mathematics (see [3] for a catalogue of fifty-two representations):

Branch	Form
Complex analysis	Gamma ratio $\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$
Classical geometry	Scaled lemniscate arc length $2\varpi/\sqrt{\pi}$
Elliptic integrals	$2\sqrt{2} K(1/\sqrt{2})/\sqrt{\pi}$
Computational number theory	$2\sqrt{\pi}/\text{agm}(1, \sqrt{2})$
Modular forms	$\sqrt{2\pi} \theta_3(e^{-\pi})^2$
Lattice combinatorics	$\sqrt{2\pi} \overline{W}_3$ (Watson integral)
Arithmetic geometry	$8 L(E, 1)/\sqrt{\pi}$ (BSD formula)

No other unnamed constant connects this many domains. The constant π is the product side of the reflection formula; G^* is the ratio side. Both are outputs of the same algebraic identity at the same evaluation point. One was elevated to the most famous number in mathematics. The other was left on the floor.

The oversight is understandable. The product gives π , which has obvious geometric meaning (circumference, area, periodicity). The ratio gives G^* , which has no simple geometric interpretation as a length or an area. But G^* has *arithmetic* meaning: it is

the limit of the Wallis product over the Fermat residue classes, the scaled period of the simplest CM curve, the bridge between the lattice and the circle.

The question is not whether G^* is new—it is not. The question is whether it is fundamental. We believe the evidence presented here, and in the companion papers [3, 4], answers in the affirmative.

8. A NOTE ON NAMING

We have used the notation G^* throughout, following its role as the “bridge constant” in Foundational Ternary Dynamics. We note that the constant admits a simpler description than any notation suggests:

G^ is the ratio of the Gamma function at the quarter-point to its reflection.*

A dedicated symbol is warranted by the same criterion that warranted π and ϖ : the constant appears independently in enough contexts that writing it out in full each time obscures the structure it reveals.

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